

Alisha Chang

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EDUCATION

Barnard College of Columbia University

New York, NY

BA in Computer Science, Minor in Economics | GPA: 3.62, Major GPA: 3.74

Sep. 2025 – May 2028

Washington University in St. Louis

St. Louis, MO

BA in Computer Science and Economics, Minor in Psychology | GPA: 4.0, Major GPA: 4.0

Aug. 2024 – May 2025

EXPERIENCE

Research Assistant

Jan. 2026 – Present

Columbia University Medical Center and Programming Languages Lab

New York, NY

- Enhanced Spiral Analysis, a React/Next.js web platform that captures tablet-drawn spirals and sends coordinate, pressure, and timestamp data to a MATLAB-based API to quantify motor symptoms through DOS, tremor, tightness, smoothness, and COV-of-width metrics.
- Led usability testing with 12 neurologists, achieving a mean System Usability Scale (SUS) score of 75.7 (above the 68-point industry average), and refined the interface based on clinician feedback.
- Engineered the tablet data pipeline with iPad/Wacom calibration, device-pixel-ratio scaling, 10 cm canvas normalization, duplicate timestamp cleanup, and API replay scripts to improve input reliability and debugging.
- Developed Supabase-backed result storage and interactive dashboards for longitudinal left/right-hand analysis, including trend charts, filtered assessment history, and clinical metric visualizations.

Software Engineering Intern

May 2026 – June 2026

New York Department of Environmental Protection

New York, NY

- Environmental Asset Database, a Next.js fullstack web platform enabling the NYC Department of Environmental Protection to upload, store, and manage environmental asset files.
- Designed a Supabase relational database with 7 interconnected tables to model environmental asset records, researching diverse data formats to structure a schema supporting the department's asset-tracking needs.
- Engineered a FastAPI backend with Pydantic-enforced type validation across all 7 tables, ensuring data integrity for dozens of fields, and built Python pipelines for automated retrieval and cleaning to streamline multi-value lookups for staff.

PROJECTS

Redstone Door Tutor | *Bootstrap, HTML, CSS, JavaScript, Git*

Feb. 2026 – May. 2026

- Identified low student engagement in Fundamentals of Computer Systems, a notoriously difficult required course, and scoped a Minecraft-themed learning platform to teach core systems concepts through a format students already enjoyed.
- Conducted user interviews with 12 students across iterations, using feedback to prioritize increased interactivity and simplify Minecraft-specific mechanics so non-players could learn effectively.
- Collaborated with a team of 3 to design, build, and iterate on the platform end-to-end.

Sylly | *Python, Adobe PDF Services API, Cohere API*

Oct. 2025 – Nov. 2025

- Designed an end-to-end parsing pipeline using Adobe PDF Services API and Cohere NLP to extract and structure assignments and exam deadlines from unstructured PDF syllabi.
- Evaluated 5 LLM APIs and selected Cohere based on memory efficiency and response latency, demonstrating systematic model comparison methodology relevant to AI/ML decision-making.
- Reduced NLP input size by ~40–60% through targeted text cleaning, improving model performance and resource usage.

TECHNICAL SKILLS

Languages: Python, R, Java, C, Javascript, HTML, CSS

Libraries & Tools: Git/GitHub, Docker, Linux/Unix, VS Code, Supabase, Cohere API, Adobe PDF Services API

Framework: Flask, Bootstrap